

## \\MPIEA\Dimo\3DEPI\_test\3DEPI\_final\ldzne\_ep3d\_final

TA: 7:12 min Coil Selection: Manual Voxel Size: 0.4×0.4×0.4 mm<sup>3</sup> Acc:: 2 Rel. SNR: 1.00**Properties**

|                                               |                    |
|-----------------------------------------------|--------------------|
| Start measurement without further preparation | On                 |
| Wait for User to Start                        | Off                |
| Start measurements                            | Single Measurement |
| Prio Recon                                    | Off                |
| Auto Open Inline Display                      | Off                |
| Auto Close Inline Display                     | Off                |
| Load Images to MR View&GO                     | On                 |
| Auto Store Images                             | On                 |
| Disable auto transfer to PACS                 | Off                |
| Load Images to Stamp Segments                 | Off                |
| Load Images to Graphic Segments               | Off                |
| Graphic segment                               | Default            |
| Inline Movie                                  | Off                |

**Routine**

|                     |                    |
|---------------------|--------------------|
| Slab Group          | 1                  |
| Slabs               | 1                  |
| Position            | L3.8 A9.7 H24.5 mm |
| Orientation         | T > C-13.7         |
| Phase Encoding Dir. | A >> P             |
| Slices per Slab     | 416                |
| Phase Oversampling  | 0.0 %              |
| Slice Oversampling  | 3.8 %              |
| FOV Read            | 175 mm             |
| FOV Phase           | 120.0 %            |
| Slice Thickness     | 0.35 mm            |
| TR                  | 138.7 ms           |
| Vol. TR             | 419.42880 s        |
| TE 1                | 56.00 ms           |
| Averages            | 1                  |
| Multi-echo Shots    | 1                  |
| AutoAlign           | ---                |
| Coil Elements       | HEA;HEP            |

**Contrast - Common**

|                    |                     |
|--------------------|---------------------|
| TR                 | 138.7 ms            |
| Vol. TR            | 419.42880 s         |
| TE 1               | 56.00 ms            |
| Multi-echo spacing | 122.30 ms           |
| MTC                | Off                 |
| Flip Angle         | 16 deg              |
| Fat-Water Contrast | Fast Fat Saturation |
| Contrasts          | 1                   |
| Reconstruction     | Magnitude           |

**Contrast - Dynamic**

|              |          |
|--------------|----------|
| Dynamic Mode | Standard |
| Measurements | 1        |
| Reordering   | Linear   |

**Resolution - Common**

|                  |         |
|------------------|---------|
| FOV Read         | 175 mm  |
| FOV Phase        | 120.0 % |
| Slice Thickness  | 0.35 mm |
| Base Resolution  | 500     |
| Phase Resolution | 100 %   |
| Slice Resolution | 100 %   |
| Interpolation    | Off     |

**Resolution - Acceleration**

|                        |              |
|------------------------|--------------|
| Acceleration Mode      | CAIPIRINHA   |
| Reference Scans        | GRE/Separate |
| Acceleration Factor PE | 2            |
| Reference Lines PE     | 36           |
| Acceleration Factor 3D | 1            |
| Reference Lines 3D     | 36           |
| Reordering Shift 3D    | 0            |
| Phase Partial Fourier  | 7/8          |
| Slice Partial Fourier  | Off          |

**Resolution - Filter**

|                       |     |
|-----------------------|-----|
| Raw Filter            | Off |
| Elliptical Filter     | Off |
| Distortion Correction | Off |
| Normalize             | Off |
| Image Filter          | Off |

**Geometry - Common**

|                     |                    |
|---------------------|--------------------|
| Slab Group          | 1                  |
| Slabs               | 1                  |
| Position            | L3.8 A9.7 H24.5 mm |
| Orientation         | T > C-13.7         |
| Phase Encoding Dir. | A >> P             |
| Slices per Slab     | 416                |
| Phase Oversampling  | 0.0 %              |
| Slice Oversampling  | 3.8 %              |
| FOV Read            | 175 mm             |
| FOV Phase           | 120.0 %            |
| Slice Thickness     | 0.35 mm            |
| TR                  | 138.7 ms           |
| Vol. TR             | 419.42880 s        |
| Multi-echo Shots    | 1                  |

**Geometry - AutoAlign**

|                     |                    |
|---------------------|--------------------|
| Slab Group          | 1                  |
| Position            | L3.8 A9.7 H24.5 mm |
| Orientation         | T > C-13.7         |
| Phase Encoding Dir. | A >> P             |
| AutoAlign           | ---                |
| Initial Position    | L3.8 A9.7 H24.5    |
| Phase               | -15.3 mm           |

**Geometry - AutoAlign**

|                     |          |
|---------------------|----------|
| Read                | -3.8 mm  |
| Shift               | 21.5 mm  |
| Initial Orientation | T > C    |
| T > C               | -13.70   |
| > S                 | 0.00     |
| Initial Rotation    | 0.00 deg |

**Geometry - Saturation**

|                 |       |
|-----------------|-------|
| Saturation Mode | Quick |
|-----------------|-------|

**Geometry - Tim Planning Suite**

|                   |      |
|-------------------|------|
| Set-n-Go Protocol | Off  |
| Table Position    | 0 mm |
| Table Position    | H    |
| Inline Composing  | Off  |

**System - Miscellaneous**

|                     |                  |
|---------------------|------------------|
| Coil Selection      | Manual           |
| Radial Sorting      | Off              |
| MSMA                | S - C - T        |
| Sagittal            | R >> L           |
| Coronal             | A >> P           |
| Transversal         | F >> H           |
| Coil Combination    | Adaptive Combine |
| Matrix Optimization | Off              |

**System - Adjustments**

|                       |          |
|-----------------------|----------|
| Adjustment Strategy   | Standard |
| B0 Shim               | Brain    |
| B1 Shim               | TrueForm |
| Adjustment Tolerance  | Auto     |
| Adjust with Body Coil | Off      |
| Confirm Frequency     | Never    |
| Assume Silicone       | Off      |

**System - Adjust Volume**

|             |                    |
|-------------|--------------------|
| Position    | L3.8 A9.7 H24.5 mm |
| Orientation | T > C-13.7         |
| Rotation    | 90.00 deg          |
| R >> L      | 175 mm             |
| A >> P      | 210 mm             |
| F >> H      | 146 mm             |
| Reset       | Off                |

**System - pTx**

|            |           |
|------------|-----------|
| B1 Shim    | TrueForm  |
| Excitation | Slab-sel. |

**System - Tx/Rx**

|                     |                |
|---------------------|----------------|
| Frequency 1H        | 123.158717 MHz |
| ? Ref. Amplitude 1H | 0.000 V        |
| Reset               | Off            |
| Correction Factor   | 1.00           |
| Image Scaling       | 1.000          |

**Sequence - Part 1**

|               |           |
|---------------|-----------|
| Sequence Name | ep v3.1   |
| Dimension     | 3D        |
| Excitation    | Slab-sel. |
| RF Pulse Type | Normal    |
| Gradient Mode | Fast      |
| Reordering    | Linear    |
| Bandwidth     | 344 Hz/Px |
| Echo Spacing  | 3.14 ms   |
| Segmentation  | 7         |
| EPI Factor    | 38        |

**Sequence - Part 2**

|              |    |
|--------------|----|
| Introduction | On |
| RF Spoiling  | On |

**Sequence - Special**

|                       |              |
|-----------------------|--------------|
| RF duration           | 1000 us      |
| RF time x BW          | 18           |
| PAT ref. FA           | 7 deg        |
| Fat sat. FA           | 110 deg      |
| T1 (Ernst FA)         | 1200 ms      |
| Invert PE             | Off          |
| Min. TE w/ PF         | On           |
| Ramp Sampling         | On           |
| Trigger per shot      | Off          |
| Noise image           | Off          |
| Relax spoilers        | On           |
| Round up Vol. TR      | Off          |
| MT flip angle         | 500 deg      |
| MT off-res.           | 2000 Hz      |
| MT RF duration        | 10240 us     |
| Custom Water Exc.     | -none-       |
| Phase Correction      | per Series   |
| Saturation RF         | per Shot     |
| EPI rise time factor  | 1.50         |
| G. spoil dephasing[1] | 0.0 pi       |
| G. spoil dephasing[2] | 2.0 pi       |
| G. spoil dephasing[3] | 0.0 pi       |
| Modify Ice Config     | On           |
| G-factor map          | Off          |
| GRAPPA Regularization | 200 /10^6    |
| Slab Scale            | 0 %          |
| RF spoil scheme       | Conventional |
| Read polarity         | Regular RO   |

**Sequence - Assistant**

|               |     |
|---------------|-----|
| SAR Assistant | Off |
|---------------|-----|

\\MPIEA\Dimo\3DEPI\_test\3DEPI\_final\ldzne\_ep3d\_final\_invertedRO

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|                                               |                    |
|-----------------------------------------------|--------------------|
| Start measurement without further preparation | On                 |
| Wait for User to Start                        | Off                |
| Start measurements                            | Single Measurement |
| Prio Recon                                    | Off                |
| Auto Open Inline Display                      | Off                |
| Auto Close Inline Display                     | Off                |
| Load Images to MR View&GO                     | On                 |
| Auto Store Images                             | On                 |
| Disable auto transfer to PACS                 | Off                |
| Load Images to Stamp Segments                 | Off                |
| Load Images to Graphic Segments               | Off                |
| Graphic segment                               | Default            |
| Inline Movie                                  | Off                |

**Routine**

|                     |                    |
|---------------------|--------------------|
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| Slabs               | 1                  |
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| Phase Oversampling  | 0.0 %              |
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| FOV Phase           | 120.0 %            |
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| TE 1                | 56.00 ms           |
| Averages            | 1                  |
| Multi-echo Shots    | 1                  |
| AutoAlign           | ---                |
| Coil Elements       | HEA;HEP            |

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|--------------------|---------------------|
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| MTC                | Off                 |
| Flip Angle         | 16 deg              |
| Fat-Water Contrast | Fast Fat Saturation |
| Contrasts          | 1                   |
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|--------------|----------|
| Dynamic Mode | Standard |
| Measurements | 1        |
| Reordering   | Linear   |

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|------------------|---------|
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| Base Resolution  | 500     |
| Phase Resolution | 100 %   |
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| Interpolation    | Off     |

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| Acceleration Factor PE | 2            |
| Reference Lines PE     | 36           |
| Acceleration Factor 3D | 1            |
| Reference Lines 3D     | 36           |
| Reordering Shift 3D    | 0            |
| Phase Partial Fourier  | 7/8          |
| Slice Partial Fourier  | Off          |

**Resolution - Filter**

|                       |     |
|-----------------------|-----|
| Raw Filter            | Off |
| Elliptical Filter     | Off |
| Distortion Correction | Off |
| Normalize             | Off |
| Image Filter          | Off |

**Geometry - Common**

|                     |                    |
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| Saturation Mode | Quick |
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**Geometry - Tim Planning Suite**

|                   |      |
|-------------------|------|
| Set-n-Go Protocol | Off  |
| Table Position    | 0 mm |
| Table Position    | H    |
| Inline Composing  | Off  |

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| Adjustment Tolerance  | Auto     |
| Adjust with Body Coil | Off      |
| Confirm Frequency     | Never    |
| Assume Silicone       | Off      |

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| Reset       | Off                |

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|            |           |
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|---------------------|----------------|
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| Reset               | Off            |
| Correction Factor   | 1.00           |
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| Gradient Mode | Fast      |
| Reordering    | Linear    |
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| Segmentation  | 7         |
| EPI Factor    | 38        |

**Sequence - Part 2**

|              |    |
|--------------|----|
| Introduction | On |
| RF Spoiling  | On |

**Sequence - Special**

|                       |              |
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| RF duration           | 1000 us      |
| RF time x BW          | 18           |
| PAT ref. FA           | 7 deg        |
| Fat sat. FA           | 110 deg      |
| T1 (Ernst FA)         | 1200 ms      |
| Invert PE             | Off          |
| Min. TE w/ PF         | On           |
| Ramp Sampling         | On           |
| Trigger per shot      | Off          |
| Noise image           | Off          |
| Relax spoilers        | On           |
| Round up Vol. TR      | Off          |
| MT flip angle         | 500 deg      |
| MT off-res.           | 2000 Hz      |
| MT RF duration        | 10240 us     |
| Custom Water Exc.     | -none-       |
| Phase Correction      | per Series   |
| Saturation RF         | per Shot     |
| EPI rise time factor  | 1.50         |
| G. spoil dephasing[1] | 0.0 pi       |
| G. spoil dephasing[2] | 2.0 pi       |
| G. spoil dephasing[3] | 0.0 pi       |
| Modify Ice Config     | On           |
| G-factor map          | Off          |
| GRAPPA Regularization | 200 /10^6    |
| Slab Scale            | 0 %          |
| RF spoil scheme       | Conventional |
| Read polarity         | Inverted RO  |

**Sequence - Assistant**

|               |     |
|---------------|-----|
| SAR Assistant | Off |
|---------------|-----|